

JECRC UNIVERSITY, JAIPUR
Session: Jan-June 2026
Subject: Mathematics-II (DMA025A)
B. Tech. II Semester I Year 2025-26
Tutorial Sheet-10 (M.M: 64)

UNIT-5: Complex Variable-Integration

CO5: The students will learn the complex variable integration.

PART – A (5*2=10)

A1: State Taylor's Theorem.

A2: State Laurent's Theorem.

A3: Give the state Cauchy's Residue Theorem.

A4: Define isolated singularity and obtain the poles of $\frac{z^3}{(z-2)(z-3)}$.

A5: Determine the poles of the function $\frac{z^2}{(z-a)(z-b)(z-c)}$.

PART – B (3*7=21)

B1: Expand $f(z) = \sin z$ in Taylor's series about $z = \pi/4$.

$$\text{Ans. } \frac{\sqrt{2}}{2} \left\{ 1 - (z - \pi/4) + \frac{(z - \pi/4)^2}{2!} - \frac{(z - \pi/4)^3}{3!} + \dots \right\}$$

B2: Determine the poles of the following function and residue at each pole $f(z) = \frac{z^2}{(z-1)^2(z+2)}$ and

hence evaluate $\int_c \frac{z^2 dz}{(z-1)^2(z+2)}$ where $c: |z| = 3$.

$$\text{Ans. } \frac{5}{9}, \frac{4}{9} \text{ and } 2\pi i$$

B3: Expand in the series the function $f(z) = \frac{1}{z^2 - 3z + 2}$ valid in the regions (i) $|z| < 1$. (ii) $|z| > 2$.

$$\text{Ans. (i) } \sum_{n=0}^{\infty} \left(1 - \frac{1}{2^{n+1}} \right) z^n \quad (ii) \sum_{n=0}^{\infty} (2^n - 1) \frac{1}{z^{n+1}}$$

PART – C (3*11=33)

C1: Find the Laurent expansion of $f(z) = \frac{1}{(z^2+1)(z+2)}$ when $|z| > 2$ and $1 < |z| < 2$.

$$\text{Ans.: } \frac{1}{5z} \sum_{n=0}^{\infty} (-1)^n \left(\frac{2}{z} \right)^n - \frac{1}{5} \left(\frac{1}{z} - \frac{2}{z^2} \right) \sum_{n=0}^{\infty} (-1)^n \frac{1}{z^{2n}}. \quad \text{And} \quad \frac{1}{10} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z}{2} \right)^n + \frac{2-z}{5z^2} \sum_{n=0}^{\infty} (-1)^n \left(\frac{1}{z^2} \right)^n.$$

C2: Expand $f(z) = \frac{z+3}{z(z^2-z-2)}$ in the powers of z where

(i) $|z| < 1$ (ii) $1 < |z| < 2$.

$$\text{Ans. (i) } \frac{-3}{2z} + \sum_{n=0}^{\infty} \left[\frac{2}{z} (-1)^n - \frac{5}{12} \left(\frac{1}{2} \right)^n \right] z^n \quad (ii) \frac{-3}{2z} + \frac{2}{3} \sum_{n=0}^{\infty} (-1)^n \frac{1}{z^{n+1}} - \frac{5}{12} \sum_{n=0}^{\infty} \left(\frac{z}{2} \right)^n$$

C3: Evaluate the integral $\int_0^{2\pi} \frac{d\theta}{5-3\cos\theta}$. **Ans. $\frac{\pi}{2}$**
